

MAJOR EXAMINATION (28-04-2026)

Digital System Design

Indian Institute of Technology, Hyderabad

Duration: 2 hours. Total marks: 100

Your answers are yours by right; copying is neither your right nor right in an exam.

Answer all the questions. No Aids allowed. All the best!

Computer memory stores every bit with precision; may your thoughts flow just as clearly and accurately...

1. Explain the fundamental difference between volatile and non-volatile memory. Why is Flash memory suitable for long-term storage, while DRAM is preferred for high-speed CPU tasks? (4 + 4)
 2. Describe the basic structure of a **floating gate transistor**. How do the processes of "**Programming**" and "**Erasing**" change the cell's electrical state to represent a binary 0 or 1? (3 + 3)
 3. What is **Multi-Level Cell (MLC)** technology, and how does it increase storage capacity? Additionally, why has the industry shifted from horizontal 2D NAND to vertical 3D NAND structures? (2 + 2 + 4)
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*Like **Boolean algebra**, keep it simple—trust your logic, choose clearly and let the right answers follow...*

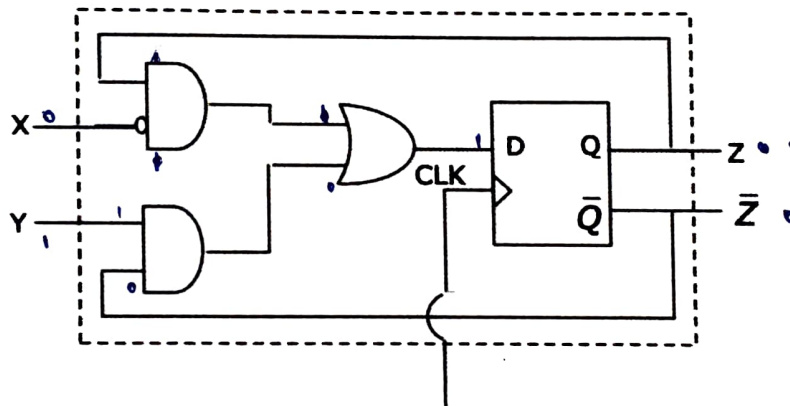
4. Write boolean expressions for sum and carry of full adder in POS form. Further in POS form find (1 + 1)
 - a. $f_1 = \text{sum and carry}$ (2)
 - b. $f_2 = \text{sum or carry}$ (2)
 - c. $f_3 = f_1\text{bar and } f_2$ (2)
 5. Realize the boolean function $f = (AB + C)D + E + A + C$ using CMOS gates. (6)
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An ADC turns real-world signals into clear digital values—just like your understanding turning effort into answers

6. An ideal **8-bit ADC** has a full-scale input range of **0 to 2 V**.
 - a. Calculate the **quantization step size (LSB value)**. (3)
 - b. Determine the **maximum quantization error**.(3)
 - c. Assuming the quantization error is uniformly distributed, compute the **quantization noise power**.(4)
 - d. What is the **Signal-to-Quantization-Noise Ratio (SQNR)** in dB for a full-scale sine wave input?(4)

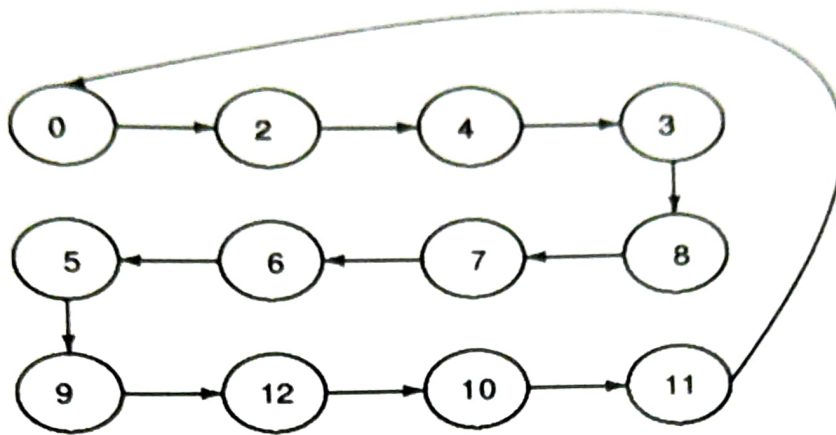
A flip-flop holds steady and changes only at the right time—stability and timing matter

7. Find the characteristic table and characteristic equation of the below diagram as a function of X and Y: (5 + 5)



A synchronous counter: everyone moves together—no one gets ahead, no one gets left behind!

8. Design a **synchronous counter** with outputs $Q_3Q_2Q_1Q_0$, where Q_3 is MSB. The counter state diagram is given as below. Use T flip-flop for Q_3 and Q_0 . Use D flip-flop for Q_2 and J-K flip-flop for Q_1 . (20)



Timing is important – Both in life and in circuits!

9. This problem involves calculating timing constraints for a sequential circuit with three registers and multiple logic paths. Setup times of all the registers are 2 ns. (4×5)

a. Determine the minimum clock period. Disregard clock skew. L1, L2, and L3 have a delay of 2ns and Mux has a delay of 5 ns. Clk to Q delay of each register is 6 ns.

b. Repeat part a, with delay of inverters as 4ns.

c. Derive the maximum delay the inverters can have that can be tolerated before the circuit fails.

d. Find the propagation delay from D to b.

